Generative AI and PolicyMap in Public Health

Instructor session plan. 90 minutes. Undergraduate students new to PolicyMap.

Learning objectives

By the end of the session students will be able to

1. Describe in plain language how generative AI produces text and why it can be fluent and wrong at the same time.
2. Apply the sliding scale of subjectivity to decide how critically to read an AI answer.
3. Write a prompt that has the AI interview them before it completes a task.
4. Generate a county level community health report in PolicyMap and have Gemini evaluate its strengths and gaps.
5. Verify AI output against the original source and locate missing data on reputable free websites.
6. Document sources in APA style, using AI to format citations without letting it invent details.

Before class

- ☐ Confirm the library's PolicyMap subscription includes the **Community Health Report**. It is a premium, county level report. If it is not included, students will use the **Community Profile** report instead, and the assignment already allows for that.
- ☐ Generate the Latah County, Idaho report yourself and save it as a PDF for your demo.
- ☐ Test uploading that PDF to Gemini with a student level campus account, and check whether the account can create a share link for a chat. This affects how students cite Gemini in Part 7.
- ☐ Post the assignment in your course site as a Word document so students can type into the boxes and upload the finished file.
- ☐ Prepare a few slides for the opening concepts, or plan to demo live in Gemini.
- ☐ Open the **+** menu under the Gemini prompt box, choose **Deep research**, and run it once with the demo prompt (Latah County adult obesity rate). Note how long it takes, whether student accounts have it, and whether its plan and revised searches are easy to show. Save screenshots in case it is slow or unavailable in class.
- ☐ Decide the due date for the reflection if students need to finish it after class.

Session at a glance

Clock	Minutes	Segment	Assignment
0:00 to 0:03	3	Welcome and objectives	Header
0:03 to 0:12	9	How generative AI works and the prediction game	Part 1a and 1b
0:12 to 0:20	8	Three principles for working with AI	Part 1c
0:20 to 0:35	15	PolicyMap demo and student reports	Part 2

Clock	Minutes	Segment	Assignment
0:35 to 0:43	8	Let Gemini interview you	Part 3
0:43 to 0:55	12	Gemini evaluates the report	Part 4
0:55 to 1:05	10	Make the AI check itself	Part 5
1:05 to 1:18	13	Finding public health data on the free web	Part 6
1:18 to 1:25	7	Documenting sources in APA style	Part 7
1:25 to 1:30	5	Wrap up and reflection	Part 8

The concepts take about 20 minutes and the hands on lab about 45, followed by 13 minutes on the free web and 7 on citation. If the lab runs long, students can finish Part 7 and Part 8 after class.

Detailed sequence

1. Welcome and objectives

Time 0:00 to 0:03 **Assignment** Header

Instructor

- Introduce yourself and the library's role in data and AI literacy.
- Share the objectives and show the assignment. Point out that the resource page at the end is theirs to keep.
- Ask students to open PolicyMap and Gemini in two browser tabs now, so login problems surface early.
- Start a **Gemini Deep Research** run on the projector before you begin. Prompt it to find the most recent adult obesity rate for Latah County, Idaho, and to show where the number comes from. Open the **+** menu under the prompt box, choose **Deep research**, and click **Start research** when its plan appears. A run takes about 10 minutes, so you will return to it at the end of segment 2. If it is still running then, show its live steps and come back to the report before class ends.

Students

- Fill in the header, open both tools, and choose a county. Suggest their home county or the county where they hope to work.

2. How generative AI works and the prediction game

Time 0:03 to 0:12 **Assignment** Part 1a and 1b

Talking points. The brain comparison

- **A neural network is loosely inspired by the brain.** An LLM is a network of simple artificial neurons joined by billions of weighted connections. Those connection weights play roughly the role of synapses.
- **Words become numbers.** Text is broken into tokens, and each token is turned into a list of numbers that places it near words it tends to appear with. Nurse sits close to hospital and patient. The associations live in the connections, not in the words themselves.

- **Training strengthens useful connections.** Each time the model guesses the next token wrong during training, its weights are nudged. Over trillions of examples the connections that predict well get stronger, much as practice strengthens synapses.
- **More connections, more capability, up to a point.** Larger models with more parameters are generally more capable. In the brain the story is more interesting. Young children have far more synapses than adults, and the brain prunes unused ones as it matures. Intelligence depends on well organized, efficient connections more than on a raw count. That makes a good bridge. Both brains and models get smarter by strengthening the right connections, not just by adding more.
- **Where the comparison breaks down.** An artificial neuron is a tiny math function, far simpler than a biological neuron. The model has no senses, no lived experience, and no way to know whether something is true. It only knows what usually comes next.

Talking points. Prediction and its limits

- **Prediction, not lookup.** The model generates answers one token at a time based on patterns, not by retrieving facts from a database.
- **Fluent is not the same as correct.** Because it predicts what sounds right, it can produce confident, well written statements that are false. These are often called hallucinations.
- **Training shapes behavior.** Companies tune models with human feedback so they are helpful, polite, and cautious. That tuning also shapes which viewpoints sound like the default.
- **The chat is its short term memory.** Everything in the conversation, including uploaded files, is what the model works from. Uploading the PolicyMap report grounds Gemini in real data.
- **Knowledge has a date.** Models have a training cutoff. Some tools search the web, but the data can still be stale.

The prediction game. Three rounds, about four minutes

- **Round 1. Familiar.** Read “Twinkle, twinkle, little star, how I wonder…” and let the class call out the ending. Nearly everyone agrees. That is a high probability prediction, like AI answering a question about glycolysis.
- **Round 2. Open ended.** Read “The biggest health problem in our community is…” and collect a few answers. They scatter. When many continuations are plausible, the answer depends on who is predicting, which previews the sliding scale of subjectivity.
- **Round 3. Familiar is not verified.** Read “I, Nephi, said unto my father…” and let the class finish it. Most will say “I will go and do the things which the Lord hath commanded.” Now ask for the exact reference and for the verse that comes right after it. Confidence drops fast. Gemini behaves the same way. It gets the famous line right because it appears everywhere in its training data, but ask it to quote the next few verses word for word and it may quietly change small words. In an October 2026 test it wrote “their tents” for “our tents” in verse 9 and dropped a word from verse 8, with full confidence. Have the official text open so the class can compare. That is a hallucination, a fluent guess where the pattern runs thin, and it is what AI does when asked for a statistic it never saw. It gets the famous numbers right and fills the gaps with plausible ones. Try this in Gemini before class so you know what it does.
- Have students jot their predictions in the Part 1a table as you go.

Not a deterministic program. About one minute

- **Show a simple math program.** A = 3, B = 5, C = A + B. Run it once or a hundred times and C is always 8. That is a deterministic program. The same input always gives the same output.
- **An LLM is different.** Ask Gemini the same question twice and the answers are similar but never exactly the same. Each word is chosen from a range of likely next words, not calculated the way C is.
- **Why it matters.** One AI answer is a sample, not the answer. That is one more reason to verify.

From predicting to problem solving. About two minutes

- **Still predicting, now in a loop.** Everything so far describes a prediction model. Newer reasoning agents still predict the next word, but they are trained to work through a problem in steps and can use tools such as web search. They plan, act, look at the result, and try something else when it does not work.
- **Three generations, one task.** Ask an early prediction model to give a county obesity rate and it writes the most likely looking answer in one pass, invented details included. A chat assistant gives a better answer in one try but does not check itself. A reasoning agent makes a plan, searches, reads what it finds, notices a dead end, changes its search terms, and tries again.
- **Picture a maze.** A prediction model draws the whole path in one guess. An agent walks it, backs up from dead ends, and tries another branch.
- **Return to the Deep Research run you started at the beginning of class.** Show its research plan and scroll through its steps so students can see it revise its searches. Point out the sources it chose, including weak ones such as YouTube or Scribd next to PubMed. Then open the report and look for something students could not trace, such as a number with no named data source or a study cited only by a registry number. Ask the class how they would check it. In an October 2026 test, Deep Research described significant results for a registered trial whose registry showed no results posted.
- **Honest caveat.** The loop makes AI more persistent, not more truthful. An agent can still misread a source or trust a weak one. Students still check every number against the PolicyMap report or the original data source.

Students

- Complete the prediction game table (1a) and write their own explanation of how Gemini produces an answer (1b).

3. Three principles for working with AI

Time 0:12 to 0:20 **Assignment** Part 1c

Principle 1. The sliding scale of subjectivity

- Refer back to rounds 1 and 2 of the prediction game. The more subjective the topic, the more critical you have to be. On glycolysis or gravity, AI will almost always give an accurate picture.
- In the middle are questions like which intervention works best or why a county's numbers look the way they do. Ask for evidence and competing explanations.
- At the subjective end are contested issues such as the Palestinian situation, vaccine mandates, or soda taxes. Treat the answer as one viewpoint among many and ask for multiple stakeholder perspectives.
- **Numbers are a special case.** Even on objective topics, specific statistics, dates, and citations are among the most common AI errors. This matters a great deal in public health.

Principle 2. Have AI verify itself

- Ask where each claim came from, what it is least sure about, and to argue the other side.
- Ask it to tie every number to a page in an uploaded document.
- **Honest caveat.** Self checks catch some errors but not all, and a model will sometimes confidently reconfirm a mistake. The final check is always a human comparing against the original source.

Principle 3. Let the AI interview you

- When a prompt is vague, the AI fills the gaps with assumptions. Flipping the roles surfaces those assumptions before the work starts.
- Template. "I want to [task]. Before you begin, interview me one question at a time, with several possible answers to choose from, until you have what you need."

Quick demo

Ask Gemini about glycolysis, then about a contested policy question. Point out how the second answer presents a framing, and ask the follow up prompt below.

> **Demo prompt for the subjective end**
>
> Give me the strongest perspectives held by different stakeholders on [ISSUE]. For each one, explain who holds it and what evidence they point to. Then tell me what kinds of sources I should read to judge these claims myself.

Students

- Add two questions of their own to the sliding scale table in Part 1c.

4. PolicyMap demo and student reports

Time 0:20 to 0:35 **Assignment** Part 2

Instructor demo with Latah County, Idaho

- Open PolicyMap from the library database list and click **Reports** at the top of the screen.
- Show the report types. The **Community Health Report** summarizes health care access, health behaviors, and social determinants and works at the county level. The **Community Profile** covers population, age, race and ethnicity, income, workforce, and transportation.
- Type Latah County, ID, generate the report, and scroll through it. Point out the data source and year printed with each indicator.
- Save as a PDF using the download or print option. If a print dialog opens, choose Save as PDF.
- **Teaching moment.** The newer Community Profile has a Generate AI Insights button that summarizes sections with an OpenAI model. AI is showing up inside the data tools themselves, and those summaries need the same scrutiny.
- **Local quirk.** Latah County is home to the University of Idaho in Moscow. A large student population can make the county look younger and poorer than its permanent residents. Ask students what quirks their own county might have.

Students

- Generate the report for their county, save the PDF, record the report type and date, and list three things that caught their attention.

5. Let Gemini interview you

Time 0:35 to 0:43 **Assignment** Part 3

Instructor

- Demo uploading the Latah PDF with the + or paperclip button, then paste the interview prompt and answer a question or two aloud.

Prompt 1. The interview

I am a public health student. I want to use the attached PolicyMap report for [YOUR COUNTY, STATE] to understand the health of this community and identify where the data falls short.

Before you do anything else, interview me. Ask me one question at a time, and give me several possible answers to choose from. Ask about my purpose, my audience, and what a useful evaluation would look like to me. When you have what you need, summarize what you learned and wait for me to say go.

Students

- Run the interview with their own report and note which question was most useful.

6. Gemini evaluates the report

Time 0:43 to 0:55 **Assignment** Part 4

Instructor

- Paste the evaluation prompt into the same chat and read part of the response aloud.
- Draw attention to the rule that forces Gemini to cite pages and label outside knowledge.
- Common gaps students may see include subgroup data by age, race, or income, mental health, substance use, maternal and child health, environmental exposures, and very local data below the county level.

Prompt 2. The evaluation

Using what you learned from our interview, evaluate the attached community health report for [YOUR COUNTY, STATE]. Organize your answer in four parts.

1. Strengths. What does this report do well? Which indicators are most useful for understanding community health?

2. Missing data. What data points would a public health professional expect in a community health assessment that are missing or thin here? Think about health outcomes, health behaviors, access to care, social determinants of health, environment, and subgroup differences.

3. Cautions. Where could these numbers mislead someone? Consider small populations, margins of error, how old the data is, and anything unusual about this county.

4. Where to look. For each missing item, name a free, reputable public source where I could find it.

Rules. Only report numbers that actually appear in the PDF and tell me the page or section where each appears. Label anything that comes from your general knowledge rather than the document as OUTSIDE THE REPORT.

Students

- Record two strengths, three missing data points, and one caution.

7. Make the AI check itself

Time 0:55 to 1:05 **Assignment** Part 5

Instructor

- Run the self check prompt and show how Gemini labels each claim.
- Model the human check by finding one of its numbers in the PDF on screen.

- Ask for a show of hands. Who found a number that did not match? Who found something Gemini flagged on its own?

> **Prompt 3. The self check**
>
> Now check your own work. List every number and factual claim you made in your last answer. For each one, tell me whether it came from the PDF (with the page or section) or from outside the report, and rate your confidence as high, medium, or low. Flag anything you cannot locate in the document, and correct any mistakes you find.

Students

- Spot check two numbers against the PDF and answer 5b.

8. Finding public health data on the free web

Time 1:05 to 1:18 **Assignment** Part 6

Talking points

- Walk through the source table on the resource page. County Health Rankings is a strong starting point because it lists the original source for every measure.
- Nonprofit hospitals must publish a community health needs assessment every three years. For Latah County, look for the Gritman Medical Center assessment and Public Health Idaho North Central District.
- Search operators. site:.gov, site:.edu, and filetype:pdf with the county name in quotes.
- Lateral reading. Before trusting a site, open a new tab and find out who runs it. The SIFT approach summarizes this as stop, investigate the source, find better coverage, and trace claims to the original.
- Always note the data year, the geography, and the method. Small counties can have large margins of error, especially in American Community Survey estimates.
- AI can suggest where to look, but students should get the number from the original source, not from the AI.

Students

- Fill one gap from Part 4 with a free source and explain why they trust it.

9. Documenting sources in APA style

Time 1:18 to 1:25 **Assignment** Part 7

Talking points

- Every number in public health work should trace back to a source someone else can check.
- AI is good at formatting and bad at remembering. When asked to recall a source, it often invents authors, dates, or DOIs. Give it the details and ask it only to format them.
- APA's current guidance for an AI chat names the company as author, gives the date of the chat, a descriptive title in italics, a bracketed description such as Google Gemini chat, the tool or model, and a share link when one exists.
- Campus Gemini accounts may not create share links. In that case APA suggests documenting the prompts, which is what the appendix box is for.

Prompt 4. Format an APA citation

Create an APA 7th edition reference list entry and a matching in text citation for the source below. Use only the details I give you. Do not invent an author, date, DOI, or web address. If something APA requires is missing, tell me what it is instead of guessing.

Source details. Title [ ]. Author or organization [ ]. Publication or update date [ ]. Web address [ ]. Date I accessed it [ ]. Type of source, for example data report, web page, or AI chat [ ].

Students

- Write APA references for the PolicyMap report, the Part 6 source, and their Gemini chat.

10. Wrap up and reflection

Time 1:25 to 1:30 **Assignment** Part 8

Key takeaways

- AI predicts. It does not look things up, so check its numbers.
- The more subjective the question, the more viewpoints you need.
- Let the AI interview you before it works for you.
- Ask the AI to verify itself, then verify it yourself.
- Cite everything, and never paste identifiable health information into an AI tool.

Students

- Start the reflection questions and submit the completed Word document by the due date.

Suggested grading

A light, completion focused rubric keeps the assignment low stress while rewarding thoughtful reflection. Twenty points total.

Section	Points	What earns full credit
Parts 1 to 3	5	Prediction game table and all boxes completed with reasonable, specific answers
Parts 4 and 5	5	Strengths, gaps, and a caution recorded. Two numbers checked against the PDF
Part 6	3	A real value from a named free source with the year and a short trust rationale
Part 7	3	Three APA references in the right pattern, with no invented details
Part 8	4	Reflection shows genuine thinking about AI in public health and appropriate use in general

Troubleshooting

Problem	What to try
Community Health Report is not listed	The subscription may not include it. Use the Community Profile. Students can still find health gaps, often more of them.
Deep Research is slow, unavailable, or still running	Show the screenshots from your practice run and come back to the live run at the end of class if it finishes.
Gemini will not accept the PDF	Check that the student is signed in with the campus account and the file is under the upload size limit. Try saving the PDF again from the print dialog.
Gemini cites numbers not in the report	That is the lesson. Have the student run the self check prompt and compare.
County has little data	Suggest a neighboring or larger county, or point to the state health department and hospital assessments.

Sources consulted

- PolicyMap Support. Making a Report. policymap.helpdocs.io/reports/making-a-report
- PolicyMap Support. New Community Profile Report. policymap.helpdocs.io/reports/new-reports
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- County Health Rankings & Roadmaps. Latah, Idaho. countyhealthrankings.org/health-data/idaho/latah
- Gritman Medical Center. Community Health Needs Assessment. gritman.org
- Public Health Idaho North Central District. idahopublichealth.com